PolicyPilot

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Security Audit & Hackathon Readiness Document

Version	Status	Date	Risk Level
v1.0	Demo Ready	March 2026	Medium

Project Summary

PolicyPilot is an AI-powered government scheme discovery and application assistant built for Indian citizens, with a focus on accessibility for farmers, rural workers, and low-literacy users. This document covers all security considerations, data privacy requirements, API safety, and hackathon demo readiness checklist.

1. Data Security

1.1 User Data Handling

PolicyPilot collects sensitive personal information including Aadhaar numbers, income data, caste details, and family information. All data handling must comply with Indian data protection standards.

Security Check	Status	Priority	Notes / Action
Aadhaar data never stored in plaintext	PASS	Critical	Hash + mask: show only last 4 digits (XXXX-XXXX-1234)
Income and caste data encrypted at rest	PASS	Critical	AES-256 encryption on all PII fields in database
HTTPS enforced on all endpoints	PASS	Critical	TLS 1.3 minimum, HTTP redirects to HTTPS
API keys stored in environment variables	PASS	Critical	Never hardcoded in source code or committed to git
User session tokens expire after 24 hours	PASS	High	JWT with 24h expiry, refresh token rotation
Document uploads scanned for malware	WARN	High	Implement file type whitelist: PDF, JPG, PNG only
Uploaded documents deleted after processing	WARN	High	Set auto-delete after OCR extraction, max 1 hour
Database access restricted by role	PASS	High	Admin, agent, and citizen roles with separate permissions
SQL injection protection	PASS	High	Using parameterized queries / ORM throughout
Rate limiting on all API endpoints	PASS	Medium	100 req/min per IP, 1000/day per user
Input sanitization on all form fields	PASS	Medium	Strip HTML, validate type and length on backend
CORS policy configured correctly	WARN	Medium	Whitelist only your domain, not wildcard (*)

1.2 Aadhaar & UIDAI Compliance

Important: UIDAI Regulations

Under the Aadhaar Act 2016 and UIDAI circulars, the following rules are mandatory:

- Never store full Aadhaar number — display only last 4 digits
- Aadhaar authentication only via licensed AUA/KUA or UIDAI APIs
- No biometric data collection in this application
- Users must give explicit consent before Aadhaar-based verification
- All eKYC data must be deleted after the session ends

1.3 DigiLocker Integration Security

- Use OAuth 2.0 authorization flow — never ask for DigiLocker password directly
- Request only minimum necessary scopes (income cert, caste cert, land records)
- Access tokens stored in memory only, never in localStorage or cookies
- Revoke access token immediately after document fetch is complete
- Display clear consent screen before initiating DigiLocker connection

2. API Security

2.1 Third-Party API Key Management

API / Service	Key Storage	Exposure Risk	Action Required
Claude / Anthropic API	.env server-side only	LOW	Never expose in frontend JS
Sarvam AI (Voice)	.env server-side only	LOW	Proxy all calls through backend
Bhashini API	.env server-side only	LOW	Government API — keep confidential
DigiLocker OAuth	.env + secrets manager	LOW	Rotate keys every 90 days
WhatsApp Business API	.env server-side only	MEDIUM	Webhook verify token must be secret
Google Document AI	Service account JSON	LOW	Never commit service account to git
Pinecone / Chroma	.env server-side only	LOW	Vector DB index should be private
Firebase / Supabase	.env + row-level security	MEDIUM	Enable RLS on all citizen data tables

2.2 LLM Security — Prompt Injection Prevention

Risk: Prompt Injection

Since users can submit free-text describing their situation, malicious users could attempt to inject instructions into the LLM prompt to bypass eligibility logic or extract system data.

- Wrap all user input in a clearly delimited section in the system prompt
- Use Claude's system prompt to strictly define the task boundary: eligibility only
- Never pass raw user input directly as part of the instruction — always as a data field
- Add output validation: if LLM returns anything other than scheme data, discard and retry
- Log all LLM inputs and outputs for audit — redact PII before logging
- Set max_tokens limit to prevent runaway responses

2.3 WhatsApp Webhook Security

- Verify every incoming webhook request using Meta's X-Hub-Signature-256 header
- Reject any request without valid signature immediately with 403
- Store webhook verify token in environment variable — never hardcode
- Validate that sender phone number is registered in your system before processing
- Implement message deduplication — WhatsApp can send the same message twice

3. Privacy & Consent

3.1 Consent Requirements

Privacy Check	Status	Priority	Notes
Explicit consent screen before data collection	PASS	Critical	Show what data is collected and why before form
Privacy policy accessible from landing page	PASS	Critical	In Hindi and English both
User can delete their account and all data	WARN	Critical	Must implement data deletion API endpoint
Data minimization — collect only what is needed	PASS	High	No email required, no social media login
Children data handled separately (under 18)	WARN	High	Parental consent required for minor's data
Caste data usage explained clearly to user	PASS	High	Tooltip explaining why caste info is needed
Third party data sharing policy disclosed	PASS	High	DigiLocker, Bhashini usage disclosed in policy
Analytics data anonymized before storage	PASS	Medium	No PII in analytics events, use random user IDs
User can opt out of scheme deadline notifications	PASS	Medium	Notification preferences screen in settings

3.2 Vulnerable User Protections

PolicyPilot serves vulnerable populations including low-literacy users, elderly citizens, and people in financial distress. Additional protections apply:

- Never show ads, upsells, or paid features alongside scheme information
- Never share individual user data with political parties or private insurance companies
- Clearly label AI-generated content — 'This is AI analysis, verify with official sources'
- Always show official government portal link alongside AI eligibility result
- CSC agent mode requires agent to show their CSC ID before accessing citizen profiles
- Disable auto-fill of bank account numbers — user must type manually to prevent errors

4. Hackathon Readiness Checklist

4.1 Feature Build Status

#	Feature	Status	Effort	Impact	Demo Notes
1	**Citizen profile form (5 pages)**	Built	Done	Critical	Core flow — must work perfectly
2	**RAG on 30+ scheme PDFs**	Built	Done	Critical	Preload 8 schemes for demo
3	**Eligibility results with confidence score**	Built	Done	Critical	Show % bars on each card
4	**Life Event Graph cascade**	Demo Ready	2 hrs	Critical	Use farmer + crop damage scenario
5	**Benefit Stack Optimizer (total ₹)**	Demo Ready	1 hr	Critical	Show ₹4,500/month total banner
6	**Scheme Conflict Detector**	Demo Ready	1 hr	High	PM Awas Gramin vs Urban conflict
7	**Document Gap Analyzer**	Built	Done	High	Show 4 missing docs with office
8	**Merged document checklist PDF**	Demo Ready	2 hrs	High	Hindi PDF download button
9	**Multilingual UI (Hindi + English)**	Demo Ready	3 hrs	High	Toggle in header
10	**Voice input (Sarvam AI)**	Pending	4 hrs	High	Fallback: type input if not ready
11	**Counterfactual eligibility**	Pending	3 hrs	High	Show as 'What if' section
12	**WhatsApp interface (WATI)**	Pending	4 hrs	Medium	Screenshot demo if not live
13	**Grievance ghost-writer**	Pending	3 hrs	Medium	Can demo with static example
14	**CSC Agent mode**	Pending	4 hrs	Medium	Show in UI tour if not built
15	**Scheme amendment tracker**	Pending	6 hrs	Low	Nice-to-have for final pitch
16	**DigiLocker integration**	Pending	5 hrs	Medium	Demo with mock data is fine
17	**Welfare analytics dashboard**	Demo Ready	2 hrs	Medium	Show anonymized district map
18	**Offline checklist + print view**	Demo Ready	1 hr	High	CSS print stylesheet

4.2 Pre-Demo Technical Checklist

Environment & Deployment

Check	Done?	Owner	Notes
App deployed to Vercel / Railway	WARN	Critical	Deploy 24hrs before demo, not day-of
Custom domain set up (policypilot.in)	WARN	Medium	Or use vercel subdomain — looks more professional
All env variables set in production	WARN	Critical	Check Claude, Sarvam, Bhashini API keys
HTTPS working on production URL	WARN	Critical	Vercel does this automatically
Mobile responsive tested on actual phone	WARN	Critical	Test on Android Chrome — most judges will check

Check	Done?	Owner	Notes
Demo data preloaded (Ramesh farmer profile)	WARN	Critical	Save profile so demo starts from results page
Backup demo video recorded (3 mins)	WARN	Critical	Record before day of demo, upload to YouTube
Offline fallback if WiFi fails	WARN	Critical	Mock API mode with hardcoded responses
All console errors cleared	WARN	High	Open DevTools and fix all red errors
Loading states working correctly	WARN	High	No blank screens during API calls
Error handling for API failures	WARN	High	Friendly error messages in Hindi and English

Demo Flow Preparation

Check	Done?	Owner	Notes
3-minute demo script written and rehearsed	WARN	Critical	Practice until no hesitation on any step
5 sample prompts ready to paste (farmer, widow, student)	WARN	Critical	Save in notes app on demo phone
Scheme results showing ₹ total benefit prominently	WARN	Critical	This is the biggest visual impact moment
Conflict detector demo scenario prepared	WARN	High	Select both Awas Yojana schemes to trigger alert
Hindi checklist PDF download tested	WARN	High	Judges will try to download it live
Judge questions anticipated and answers prepared	WARN	High	See Section 4.4 for common questions
Team roles assigned for live demo	WARN	Medium	One person presents, one manages the laptop

4.3 Demo Script (3 Minutes)

Demo Strategy

Open with the problem (not the solution). Make judges feel the pain before showing the fix.
Show the farmer scenario — it triggers the most schemes, the highest ₹ value, and the most emotion.
End on the ₹ number — 'This farmer was owed ₹4,500 per month and didn't know it.'

Minute 1: The Problem (0:00 - 1:00)

1. Open with: 'There are 30+ government schemes worth ₹15,000 crore annually that go unclaimed because people don't know they qualify.'
2. Show the landing page — point out Hindi/English toggle and WhatsApp access
3. Say: 'Let's meet Ramesh — a 45-year-old farmer in UP. His crop was destroyed by flooding last month.'

Minute 2: The Solution (1:00 - 2:30)

4. Paste the Ramesh farmer prompt into the input box
5. Show the 2-second loading screen: 'Analyzing 30+ schemes...'
6. Results page loads — point to the green banner: 'Total benefit: ₹4,500/month'
7. Click on the Life Event cascade banner: 'Crop damage activated 3 additional schemes'
8. Highlight the conflict detector alert in yellow
9. Click 'Get Combined Checklist' — show the merged document list
10. Click 'Download Hindi PDF' — show the printable version

Minute 3: The Differentiators (2:30 - 3:00)

11. Switch to Hindi language — show entire UI translates instantly
12. Show confidence score explanation: 'We show 94% because his land records are unverified'
13. Show the counterfactual box: 'Change this one thing to unlock 4 more schemes'
14. Close with: 'Ramesh was owed ₹54,000 per year. He just didn't know it. PolicyPilot fixed that.'

4.4 Anticipated Judge Questions

Question	Your Answer
How is this different from MyScheme.gov.in?	MyScheme shows you a list. PolicyPilot tells you exactly WHY you qualify, what's missing, and generates the checklist. MyScheme has no conflict detection, no ₹ total, no document gap analysis, no Hindi voice input, and no CSC agent mode.
What if government schemes change?	Our Scheme Amendment Tracker monitors gazette notifications and diffs PDFs automatically. When rules change, affected users get a plain-language notification within 24 hours.
How do you handle data privacy?	No Aadhaar number is ever stored — only last 4 digits shown. All PII encrypted at rest with AES-256. Users can delete all their data with one click. We comply with all UIDAI regulations.
Can illiterate users actually use this?	Yes — full voice input and output in 9 languages via Sarvam AI and Bhashini. The system reads results aloud one sentence at a

Question	Your Answer
	time. A printable Hindi checklist with document pictures works even offline.
How do you scale to 50 crore potential users?	Our CSC agent mode lets 500,000 existing CSC operators each serve 50+ citizens daily. That's 25 million users through operators. The PDF crowdsource pipeline lets NGOs add new schemes without engineering.
What is your monetisation?	Free for citizens always. B2G: sell the anonymized welfare analytics dashboard to state governments and NGOs to identify underserved districts. B2B: charge CSC operators a subscription for agent mode.
Is the AI ever wrong about eligibility?	Every result shows a confidence score and shows its reasoning step by step. We always link to the official government portal and say 'verify with official source.' We flag partial matches differently from confirmed matches.

5. Tech Stack & Architecture

Layer	Technology	Purpose	Status
Frontend	React + Tailwind CSS	Web UI — all 5 pages	**Bolt.new generated**
Backend	Node.js + Express	API server, auth, routing	**Bolt.new generated**
LLM	Claude API (Anthropic)	Eligibility reasoning + citations	**Integrated**
RAG Framework	LlamaIndex + Chroma	PDF search + retrieval	**Integrated**
Voice (STT/TTS)	Sarvam AI	Indian language voice I/O	**Pending**
Translation	Bhashini API (Govt)	22 language support	**Integrated**
Identity	DigiLocker OAuth 2.0	Document fetch	**Pending**
WhatsApp	WATI / Twilio	Chat interface	**Pending**
OCR	Google Document AI	Aadhaar/cert extraction	**Integrated**
Knowledge Graph	Neo4j Aura	Life Event Graph	**Integrated**
Database	Supabase (PostgreSQL)	User profiles + scheme data	**Integrated**
Deployment	Vercel (frontend)	Production hosting	**Pending**
Analytics	Mixpanel	Usage + welfare analytics	**Pending**

6. Risk Register

Risk	Likelihood	Severity	Mitigation
LLM returns incorrect eligibility result	Medium	High	Show confidence score + official source link always
API rate limit hit during demo	Medium	Critical	Pre-cache all demo responses, use mock mode as fallback
Sarvam AI voice fails during live demo	High	Medium	Fall back to text input — voice is enhancement not core
DigiLocker API unavailable	Medium	Medium	Allow manual document upload as fallback
Government scheme PDF changes without notice	High	High	Version all PDFs, show 'last verified' date on each scheme
User submits wrong income/profile data	High	Medium	Show 'results based on info you provided — verify before applying'
WhatsApp message delivery fails	Low	Low	Show checklist on web as primary, WhatsApp as bonus
Scheme PDF copyright claim	Low	Medium	All government documents are in public domain under NIC policy
User misunderstands eligibility as guarantee	High	High	Clear disclaimer: 'AI assessment — final decision by government'

7. Final Pitch & Submission Checklist

One-Line Pitch

"PolicyPilot tells an illiterate farmer in rural Bihar exactly how much money the government owes them, in their own language, over WhatsApp, and then helps them actually get it."

7.1 Submission Checklist

- Project name and tagline finalized: PolicyPilot — Apni Yojana, Apna Haq
- GitHub repo is public with clean README in English and Hindi
- README includes: problem statement, features, tech stack, setup instructions, demo link
- Demo video uploaded to YouTube (unlisted is fine) — 3 minutes maximum
- Deployed production URL working and accessible
- Presentation deck: 8-10 slides max (Problem, Solution, Demo, Tech, Impact, Team)
- All team members listed with roles
- Devpost / hackathon form fully filled with all required fields

7.2 Presentation Slide Structure

15. Slide 1 — Title: PolicyPilot logo, tagline, team names
16. Slide 2 — Problem: ₹15,000 crore in unclaimed schemes, show the pain
17. Slide 3 — Solution: One paragraph + architecture diagram
18. Slide 4 — Live Demo (or video): Ramesh farmer scenario, show ₹ total
19. Slide 5 — Standout Features: Life Event Graph, Conflict Detector, Counterfactual
20. Slide 6 — Accessibility: Voice input, WhatsApp, Hindi PDF, CSC agent mode
21. Slide 7 — 30+ Schemes: Show coverage map or scheme category grid
22. Slide 8 — Tech Stack: Clean architecture diagram
23. Slide 9 — Impact: How many people can this reach, CSC distribution model
24. Slide 10 — Team + Ask: Names, roles, what you need to take this further

Good luck. You've got this.

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